

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built a cloud-hosted data pipeline on **AWS** that indexes **600K+ documents** on *Elasticsearch* via *FastAPI*, returning results in **under 1 second**, serving **10K+ site visitors**, as part of an *Agile* team of **10 engineers**
- Improved data quality with a de-duplication algorithm that caught **30%** of evidence documents as identical
- Validated source URLs and filtered statistical outliers with **IQR** in a text-retrieval algorithm for data quality
- Tuned result ordering to favor recent evidence, putting the current month's debate topic first in the search results

Product Development Intern | Jetson

January 2025 – February 2025

- Tested and validated **3 experimental features** through **7 iterations of UI/UX testing** on the Jetson App
- Generated insights from testing in a **5-page report** recommending strategies to optimize user engagement
- Condensed user onboarding to **5 pages** from **12** in *Figma*, then **A/B tested** the new flow on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Investigated the **3 years** of TikTok data spanning **200+ posts** using *Python (pandas)*, *Power BI* and *Excel*
- Identified a trend in the data: "science fun facts" posts comprised **69.3%** of views from just **7%** of content
- Generated an insight that the Executive Director adopted as a "fun facts" strategy, lifting engagement **24%**

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Automated paper ingestion via *OpenAlex*, backfilling gaps with *Semantic Scholar* metadata in a dual-API pipeline
- Filtered papers by field, year and citation count with *SQL* queries on a relational *Supabase (PostgreSQL)* database
- Visualized a paper's outbound citations across **26 research fields** in an interactive *React Flow* citation graph
- Deployed a **RAG** generative AI agent (*Claude Sonnet 4.5*) that retrieves session-scoped unstructured documents

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Built a **15,000-image** truck-listing dataset through scraping, vetting it with de-duplication and non-USD filtering
- Transformed unstructured truck photos into structured condition, brand, model and year data using a *JSON schema* and a *Claude Sonnet 4.6* agent, validating it until **10 consecutive images** were correct on each of **3 features**
- Priced trucks with *quantile regression* on the agent's extracted features, hitting **87% accuracy** on **30 samples**
- Directed a team of **4** on **Kamion (YC S22)**'s pricing algorithm, winning **2nd at UIUC Founder's Hackathon**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Ran **4 Random Forest** machine learning models in a *Python FastAPI* back end on live data ingested from a school's **3 tools** to detect anomalies and deliver actionable steps via a *Gemma* Voice LLM on a *Next.js* dashboard
- Prototyped and built a **suite of 3 edge AI tools** that monitor a school's energy, water and food consumption
- Steered a team of **5** from ideation to presentation, earning a **USAII Global Hackathon Finalist** placement

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Achieved **96.13% accuracy** classifying waste items into 4 disposal pathways with a two-stage *EfficientNetV2-S* deep learning network (**4.39M** parameters) trained in *PyTorch* on **15K+ images**, then published a paper on *arXiv*
- Automated a synthetic data pipeline using *U2-Net* segmentation and alpha-compositing that produced **9K** images
- Placed **2nd at the Arizona State Science Fair** and posted **2 original datasets** on *Kaggle* for the project

SKILLS

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

Frameworks/Platforms: PostgreSQL, MySQL, Supabase, Elasticsearch, AWS, Google Cloud, FastAPI, Flask

AI/ML: scikit-learn, PyTorch, TensorFlow, Keras, Hugging Face, Claude API, OpenCV, Gemma 4

Data & Analytics: pandas, NumPy, Power BI, Microsoft Excel, A/B Testing, Hootsuite

Concepts/Practices: Data Integration, Data Preparation, Data Transformation, Data Validation, Data Quality